

Website_Traffic_Analysis

September 21, 2026

```
[4]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]: df = pd.read_csv("data-export (1).csv")
```

```
[3]: df.head()
```

```
[3]: # ----- \
0 Session primary channel group (Default channel...
1                                     Direct
2                               Organic Social
3                                     Direct
4                               Organic Social

      Unnamed: 1 Unnamed: 2 Unnamed: 3      Unnamed: 4 \
0 Date + hour (YYYYMMDDHH)      Users  Sessions Engaged sessions
1      2024041623             237      300             144
2      2024041719             208      267             132
3      2024041723             188      233             115
4      2024041718             187      256             125

      Unnamed: 5      Unnamed: 6 \
0 Average engagement time per session Engaged sessions per user
1      47.526666666666700      0.6075949367088610
2      32.09737827715360      0.6346153846153850
3      39.93991416309010      0.6117021276595740
4      32.16015625      0.6684491978609630

      Unnamed: 7      Unnamed: 8      Unnamed: 9
0 Events per session      Engagement rate      Event count
1      4.673333333333330      0.48      1402
2      4.295880149812730      0.4943820224719100      1147
3      4.587982832618030      0.49356223175965700      1069
4      4.078125      0.48828125      1044
```

```
[5]: df.columns = df.iloc[0]
df = df.drop(index = 0).reset_index(drop = True)
df.columns = ["Channel group", "Datehour", "User", "Sessions", "Engaged_
↳sessions", "Average engagement time per session", "Engaged sessions per_
↳user", "Events per session", "Engagement rate", "Event count"]
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[5], line 1
----> 1 df.columns = df.iloc[0]
      2 df = df.drop(index = 0).reset_index(drop = True)
      3 df.columns = ["Channel group", "Datehour", "User", "Sessions", "Engaged_
↳sessions", "Average engagement time per session", "Engaged sessions per_
↳user", "Events per session", "Engagement rate", "Event count"]

NameError: name 'df' is not defined
```

```
[6]: import pandas as pd
df = pd.read_csv("data-export (1).csv")
```

```
[7]: df.columns = df.iloc[0]
df = df.drop(index = 0).reset_index(drop = True)
df.columns = ["Channel group", "Datehour", "User", "Sessions", "Engaged_
↳sessions", "Average engagement time per session", "Engaged sessions per_
↳user", "Events per session", "Engagement rate", "Event count"]
```

```
[8]: df.head()
```

```
[8]:
```

	Channel group	Datehour	User	Sessions	Engaged sessions \
0	Direct	2024041623	237	300	144
1	Organic Social	2024041719	208	267	132
2	Direct	2024041723	188	233	115
3	Organic Social	2024041718	187	256	125
4	Organic Social	2024041720	175	221	112

	Average engagement time per session	Engaged sessions per user \
0	47.526666666666700	0.6075949367088610
1	32.09737827715360	0.6346153846153850
2	39.93991416309010	0.6117021276595740
3	32.16015625	0.6684491978609630
4	46.918552036199100	0.64

	Events per session	Engagement rate	Event count
0	4.6733333333333330	0.48	1402
1	4.295880149812730	0.4943820224719100	1147
2	4.587982832618030	0.49356223175965700	1069

3	4.078125	0.48828125	1044
4	4.529411764705880	0.5067873303167420	1001

```
[9]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3182 entries, 0 to 3181
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Channel group                        3182 non-null   object
1   Datehour                            3182 non-null   object
2   User                                3182 non-null   object
3   Sessions                            3182 non-null   object
4   Engaged sessions                    3182 non-null   object
5   Average engagement time per session 3182 non-null   object
6   Engaged sessions per user            3182 non-null   object
7   Events per session                   3182 non-null   object
8   Engagement rate                      3182 non-null   object
9   Event count                         3182 non-null   object
dtypes: object(10)
memory usage: 248.7+ KB
```

```
[14]: df["Datehour"] = pd.
      ↪to_datetime(df["Datehour"],format="%Y%m%d%H",errors='coerce')
```

```
[15]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3182 entries, 0 to 3181
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Channel group                        3182 non-null   object
1   Datehour                            3182 non-null   datetime64[ns]
2   User                                3182 non-null   object
3   Sessions                            3182 non-null   object
4   Engaged sessions                    3182 non-null   object
5   Average engagement time per session 3182 non-null   object
6   Engaged sessions per user            3182 non-null   object
7   Events per session                   3182 non-null   object
8   Engagement rate                      3182 non-null   object
9   Event count                         3182 non-null   object
dtypes: datetime64[ns](1), object(9)
memory usage: 248.7+ KB
```

```
[18]: numeric_cols = df.columns.drop(["Channel group", "Datehour"])
df[numeric_cols] = df[numeric_cols].apply(pd.to_numeric, errors='coerce')
df["Hour"] = df["Datehour"].dt.hour
```

```
[19]: df.head()
```

```
[19]:
```

	Channel group	Datehour	User	Sessions	Engaged sessions \
0	Direct	2024-04-16 23:00:00	237	300	144
1	Organic Social	2024-04-17 19:00:00	208	267	132
2	Direct	2024-04-17 23:00:00	188	233	115
3	Organic Social	2024-04-17 18:00:00	187	256	125
4	Organic Social	2024-04-17 20:00:00	175	221	112

	Average engagement time per session	Engaged sessions per user \
0	47.526667	0.607595
1	32.097378	0.634615
2	39.939914	0.611702
3	32.160156	0.668449
4	46.918552	0.640000

	Events per session	Engagement rate	Event count	Hour
0	4.673333	0.480000	1402	23
1	4.295880	0.494382	1147	19
2	4.587983	0.493562	1069	23
3	4.078125	0.488281	1044	18
4	4.529412	0.506787	1001	20

```
[20]: df.describe()
```

```
[20]:
```

	Datehour	User	Sessions \
count	3182	3182.000000	3182.000000
mean	2024-04-20 01:17:07.278441216	41.935889	51.192646
min	2024-04-06 00:00:00	0.000000	1.000000
25%	2024-04-13 02:15:00	20.000000	24.000000
50%	2024-04-20 02:00:00	42.000000	51.000000
75%	2024-04-26 22:00:00	60.000000	71.000000
max	2024-05-03 23:00:00	237.000000	300.000000
std	NaN	29.582258	36.919962

	Engaged sessions	Average engagement time per session \
count	3182.000000	3182.000000
mean	28.325581	66.644581
min	0.000000	0.000000
25%	13.000000	32.103034
50%	27.000000	49.020202
75%	41.000000	71.487069
max	144.000000	4525.000000

std	20.650569	127.200659
-----	-----------	------------

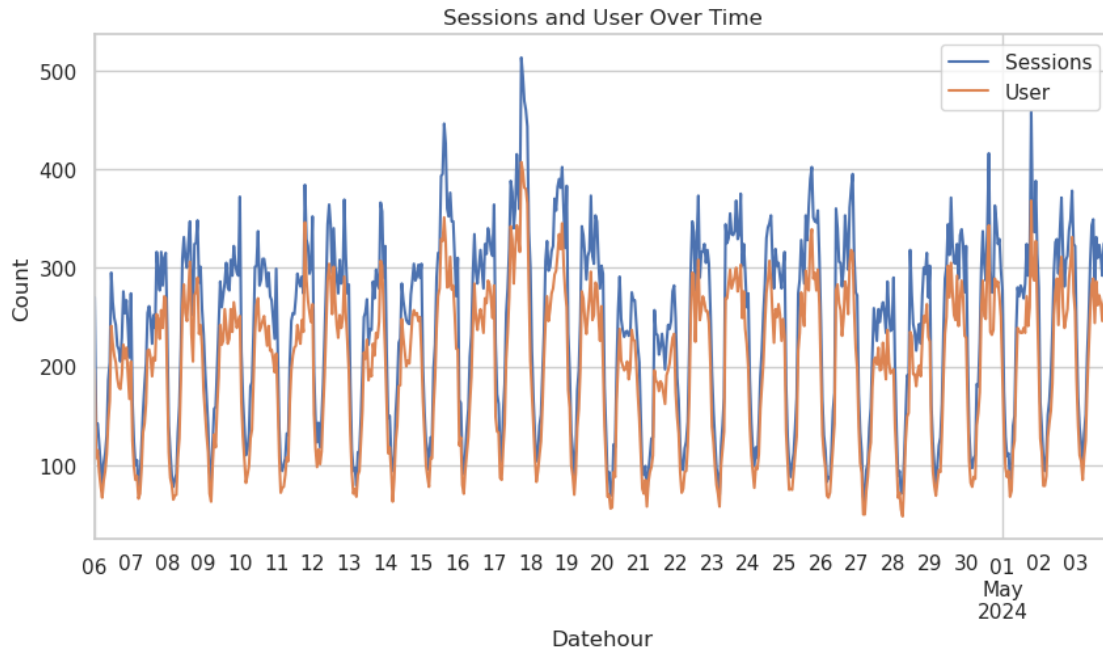
	Engaged sessions per user	Events per session	Engagement rate \
count	3182.000000	3182.000000	3182.000000
mean	0.606450	4.675969	0.503396
min	0.000000	1.000000	0.000000
25%	0.561404	3.750000	0.442902
50%	0.666667	4.410256	0.545455
75%	0.750000	5.217690	0.633333
max	2.000000	56.000000	1.000000
std	0.264023	2.795228	0.228206

	Event count	Hour
count	3182.000000	3182.000000
mean	242.272470	11.807040
min	1.000000	0.000000
25%	103.000000	6.000000
50%	226.000000	12.000000
75%	339.000000	18.000000
max	1402.000000	23.000000
std	184.440313	6.886686

1 Sessions and User Time

```
[24]: sns.set(style="whitegrid")
```

```
[30]: plt.figure(figsize=(10,5))
df.groupby("Datehour")[["Sessions","User"]].sum().plot(ax=plt.gca())
plt.title("Sessions and User Over Time")
plt.xlabel("Datehour")
plt.ylabel("Count")
plt.show()
```



2 Total Users By Channel

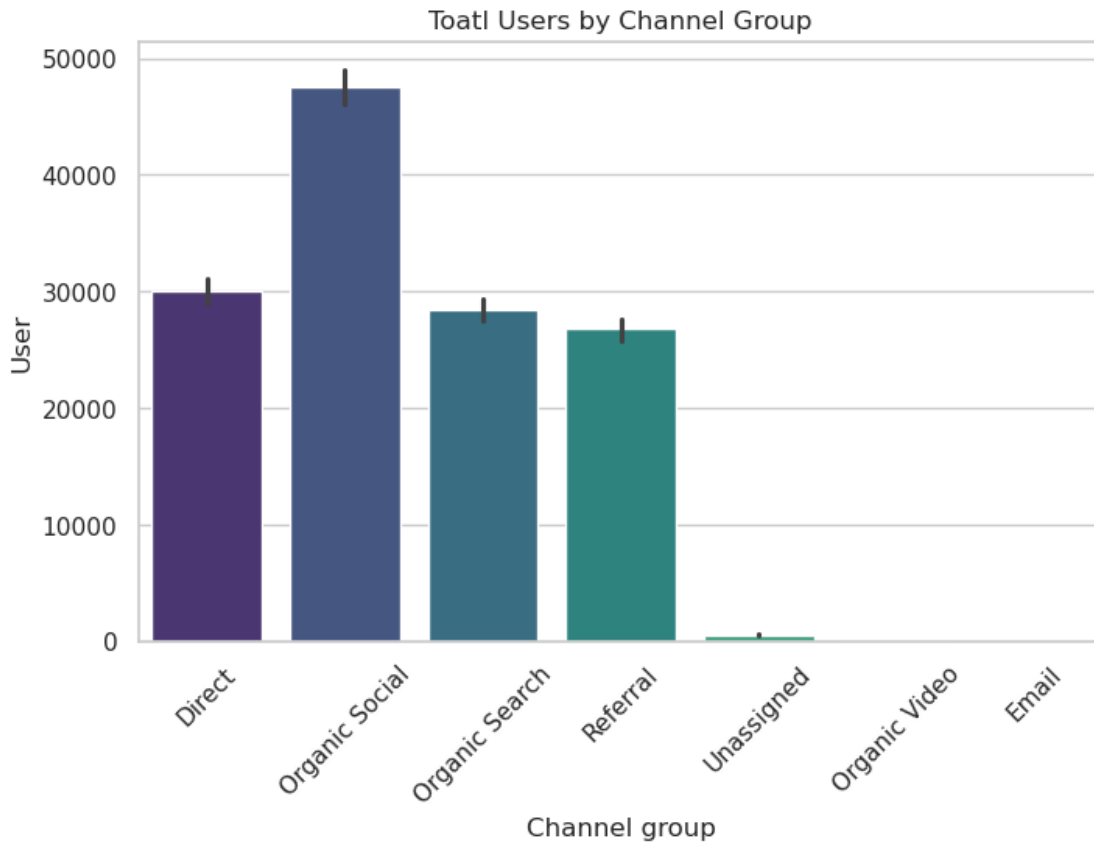
```
[33]: plt.figure(figsize=(8,5))
plt.figure(figsize=(8,5))
sns.barplot(data=df, x="Channel group", y="User", estimator=np.sum,
            palette="viridis")
plt.title("Toatl Users by Channel Group")
plt.xticks(rotation=45)
plt.show()
```

/tmp/ipykernel_127/1562703303.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=df, x="Channel group", y="User", estimator=np.sum,
palette="viridis")
```

<Figure size 800x500 with 0 Axes>



```
[34]: df.head()
```

```
[34]:
```

	Channel group	Datehour	User	Sessions	Engaged sessions \
0	Direct	2024-04-16 23:00:00	237	300	144
1	Organic Social	2024-04-17 19:00:00	208	267	132
2	Direct	2024-04-17 23:00:00	188	233	115
3	Organic Social	2024-04-17 18:00:00	187	256	125
4	Organic Social	2024-04-17 20:00:00	175	221	112

	Average engagement time per session	Engaged sessions per user \
0	47.526667	0.607595
1	32.097378	0.634615
2	39.939914	0.611702
3	32.160156	0.668449
4	46.918552	0.640000

	Events per session	Engagement rate	Event count	Hour
0	4.673333	0.480000	1402	23
1	4.295880	0.494382	1147	19
2	4.587983	0.493562	1069	23

3	4.078125	0.488281	1044	18
4	4.529412	0.506787	1001	20

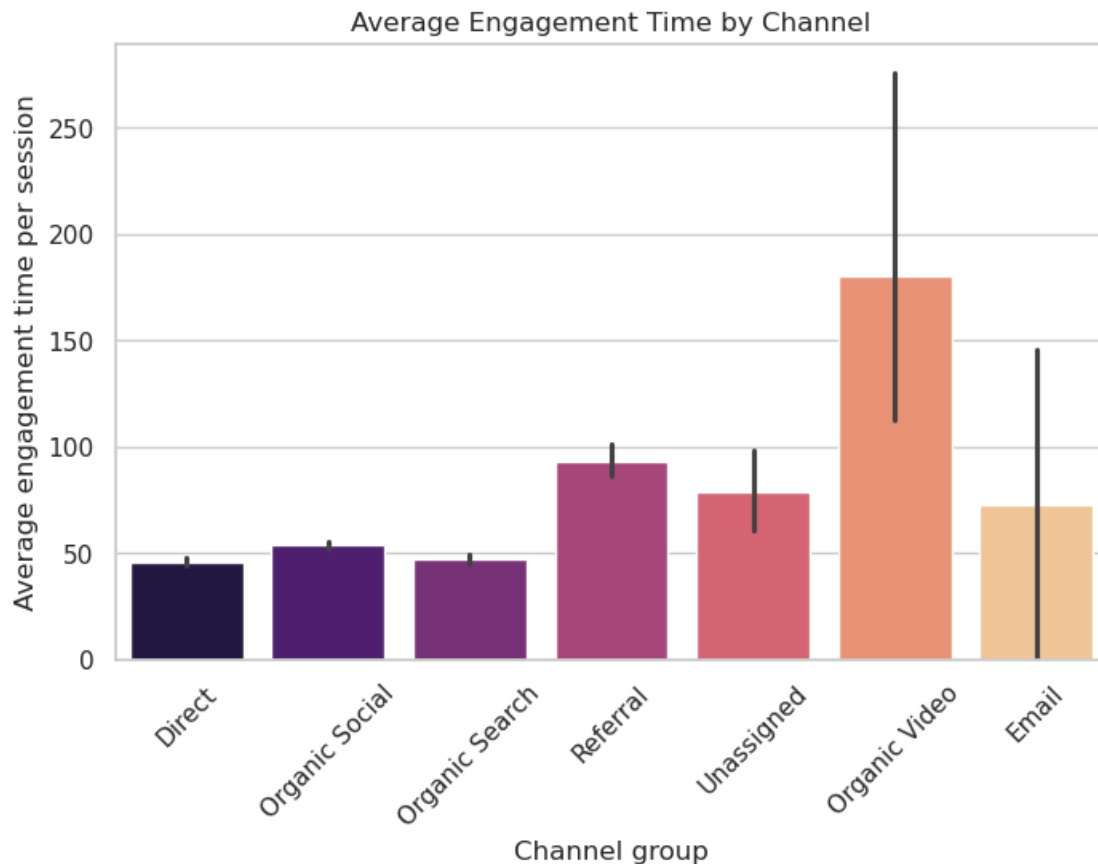
3 Average Engagement Time by Channel

```
[36]: plt.figure(figsize=(8,5))
sns.barplot(data=df, x="Channel group", y="Average engagement time per_
↪session", estimator=np.mean, palette="magma")
plt.title("Average Engagement Time by Channel")
plt.xticks(rotation=45)
plt.show()
```

/tmp/ipykernel_127/3236115117.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=df, x="Channel group", y="Average engagement time per
session", estimator=np.mean, palette="magma")
```



```
[37]: df.head()
```

```
[37]:
```

	Channel group	Datehour	User	Sessions	Engaged sessions \
0	Direct	2024-04-16 23:00:00	237	300	144
1	Organic Social	2024-04-17 19:00:00	208	267	132
2	Direct	2024-04-17 23:00:00	188	233	115
3	Organic Social	2024-04-17 18:00:00	187	256	125
4	Organic Social	2024-04-17 20:00:00	175	221	112

	Average engagement time per session	Engaged sessions per user \
0	47.526667	0.607595
1	32.097378	0.634615
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4	46.918552	0.640000

	Events per session	Engagement rate	Event count	Hour
0	4.673333	0.480000	1402	23
1	4.295880	0.494382	1147	19
2	4.587983	0.493562	1069	23
3	4.078125	0.488281	1044	18
4	4.529412	0.506787	1001	20

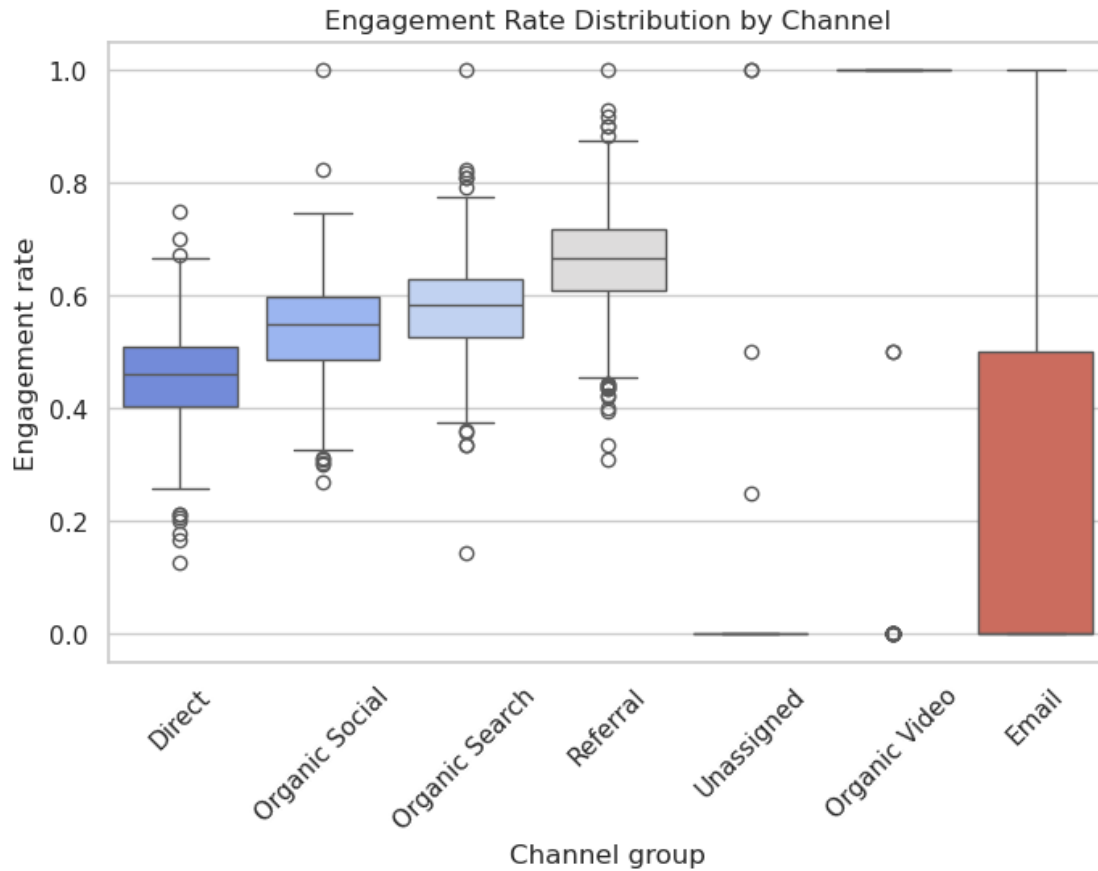
4 Engagement Rate Distribution by Channel

```
[38]: plt.figure(figsize=(8,5))
sns.boxplot(data=df, x="Channel group", y="Engagement rate", palette="coolwarm")
plt.title("Engagement Rate Distribution by Channel")
plt.xticks(rotation=45)
plt.show()
```

/tmp/ipykernel_127/3423311072.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df, x="Channel group", y="Engagement rate",
palette="coolwarm")
```



5 Engaged v/s Non-Engaged Sessions

```
[39]: df.head()
```

```
[39]:
```

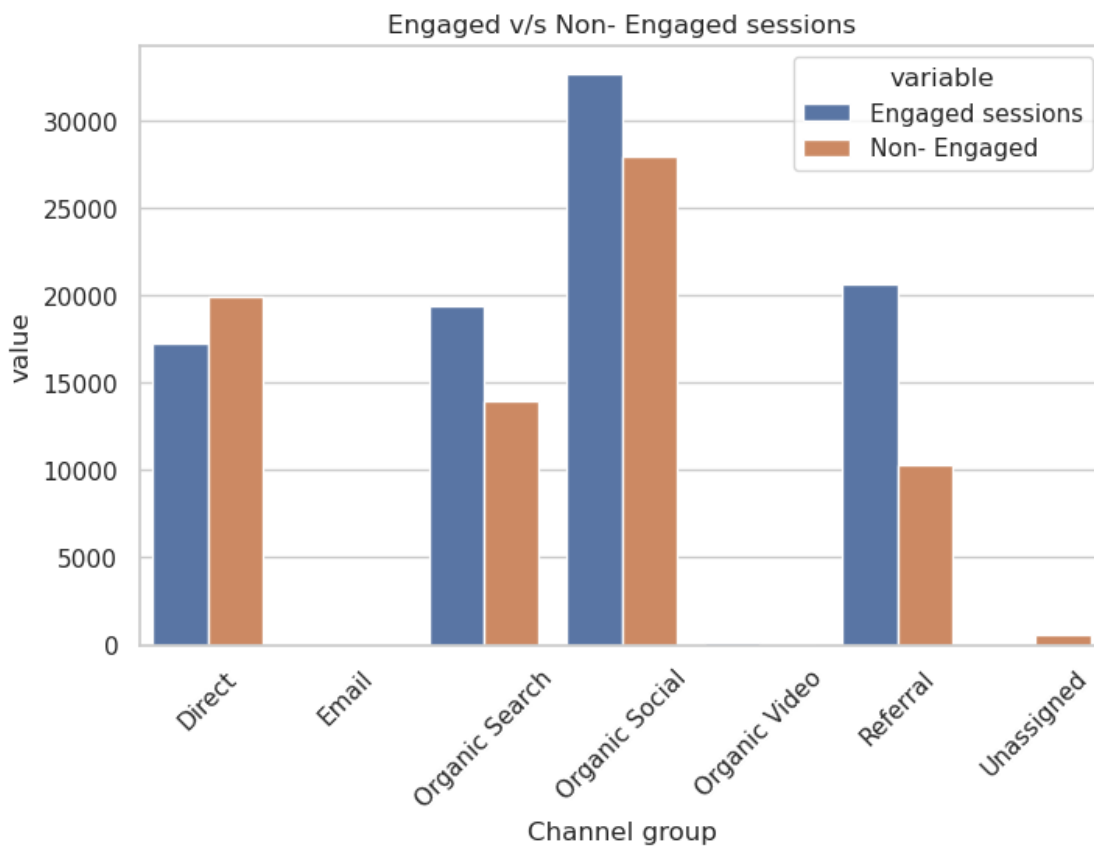
	Channel group	Datehour	User	Sessions	Engaged sessions \
0	Direct	2024-04-16 23:00:00	237	300	144
1	Organic Social	2024-04-17 19:00:00	208	267	132
2	Direct	2024-04-17 23:00:00	188	233	115
3	Organic Social	2024-04-17 18:00:00	187	256	125
4	Organic Social	2024-04-17 20:00:00	175	221	112

	Average engagement time per session	Engaged sessions per user \
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3	32.160156	0.668449
4	46.918552	0.640000

	Events per session	Engagement rate	Event count	Hour
0	4.673333	0.480000	1402	23
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2	4.587983	0.493562	1069	23
3	4.078125	0.488281	1044	18
4	4.529412	0.506787	1001	20

```
[42]: session_df = df.groupby("Channel group")[["Sessions" , "Engaged sessions"]].
      ↪sum().reset_index()
session_df["Non- Engaged"] = session_df["Sessions"] - session_df["Engaged_
      ↪sessions"]
session_df_melted = session_df.melt(id_vars="Channel_
      ↪group",value_vars=["Engaged sessions","Non- Engaged"])

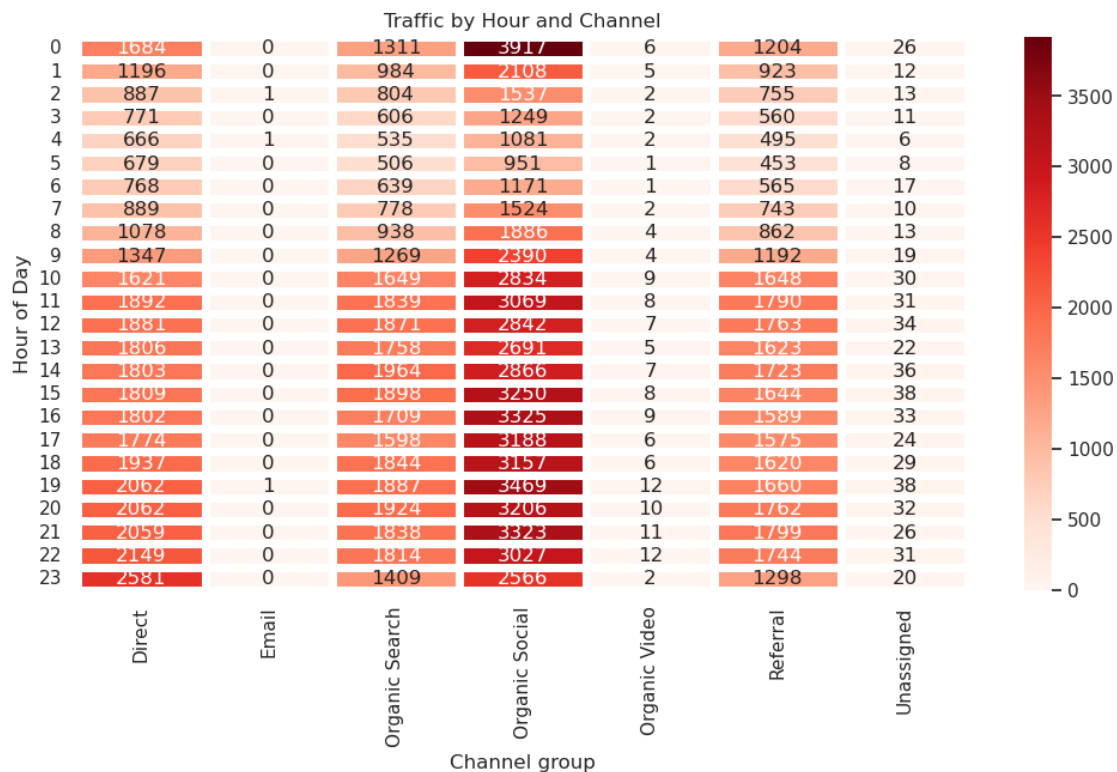
plt.figure(figsize=(8,5))
sns.barplot(data=session_df_melted, x="Channel group", y="value",
      ↪hue="variable")
plt.title("Engaged v/s Non- Engaged sessions")
plt.xticks(rotation=45)
plt.show()
```



6 Traffic by Hour and Channel

```
[47]: heatmap_data = df.groupby(["Hour", "Channel group"])["Sessions"].sum().unstack().
      ↪ fillna(0)

plt.figure(figsize=(12,6))
sns.heatmap(heatmap_data, cmap='Reds', linewidths = 5, annot=True, fmt='.0f')
plt.title("Traffic by Hour and Channel")
plt.xlabel("Channel group")
plt.ylabel("Hour of Day")
plt.show()
```



7 Engagement Rate v/s Sessions over Time

```
[48]: df.head()
```

```
[48]:
```

	Channel group	Datehour	User	Sessions	Engaged sessions \
0	Direct	2024-04-16 23:00:00	237	300	144
1	Organic Social	2024-04-17 19:00:00	208	267	132
2	Direct	2024-04-17 23:00:00	188	233	115
3	Organic Social	2024-04-17 18:00:00	187	256	125

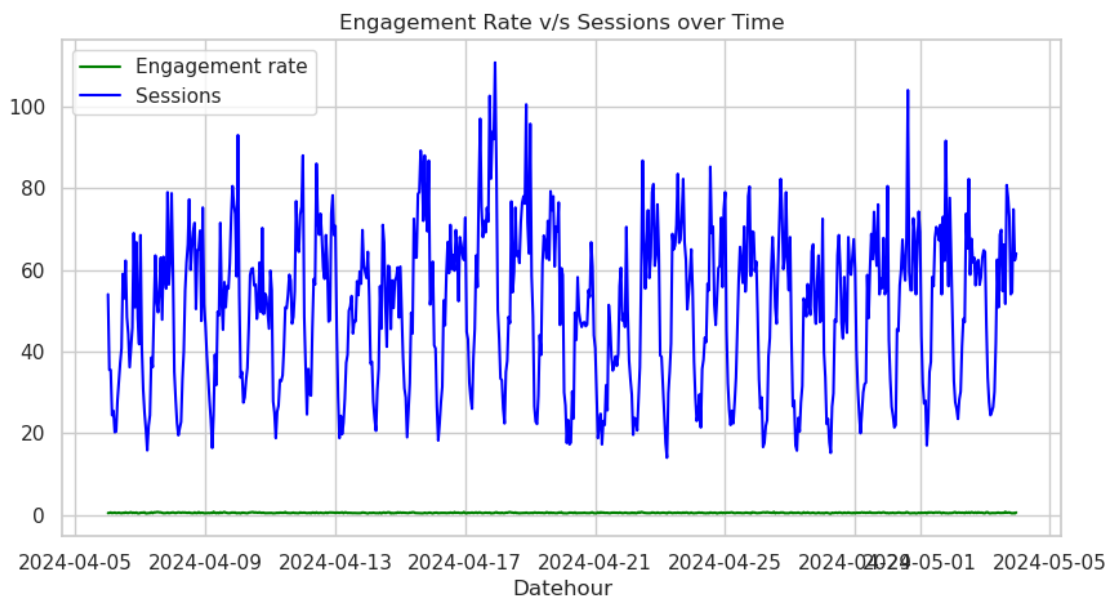
4	Organic Social	2024-04-17 20:00:00	175	221	112
---	----------------	---------------------	-----	-----	-----

	Average engagement time per session	Engaged sessions per user \
0	47.526667	0.607595
1	32.097378	0.634615
2	39.939914	0.611702
3	32.160156	0.668449
4	46.918552	0.640000

	Events per session	Engagement rate	Event count	Hour
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1	4.295880	0.494382	1147	19
2	4.587983	0.493562	1069	23
3	4.078125	0.488281	1044	18
4	4.529412	0.506787	1001	20

```
[49]: df_plot = df.groupby("Datehour")[["Engagement rate", "Sessions"]].mean().
      ↪reset_index()

plt.figure(figsize=(10,5))
plt.plot(df_plot["Datehour"], df_plot["Engagement rate"], label="Engagement_
      ↪rate",color="green")
plt.plot(df_plot["Datehour"], df_plot["Sessions"],
      ↪label="Sessions",color="blue")
plt.title("Engagement Rate v/s Sessions over Time")
plt.xlabel("Datehour")
plt.legend()
plt.grid(True)
plt.show()
```



[]: